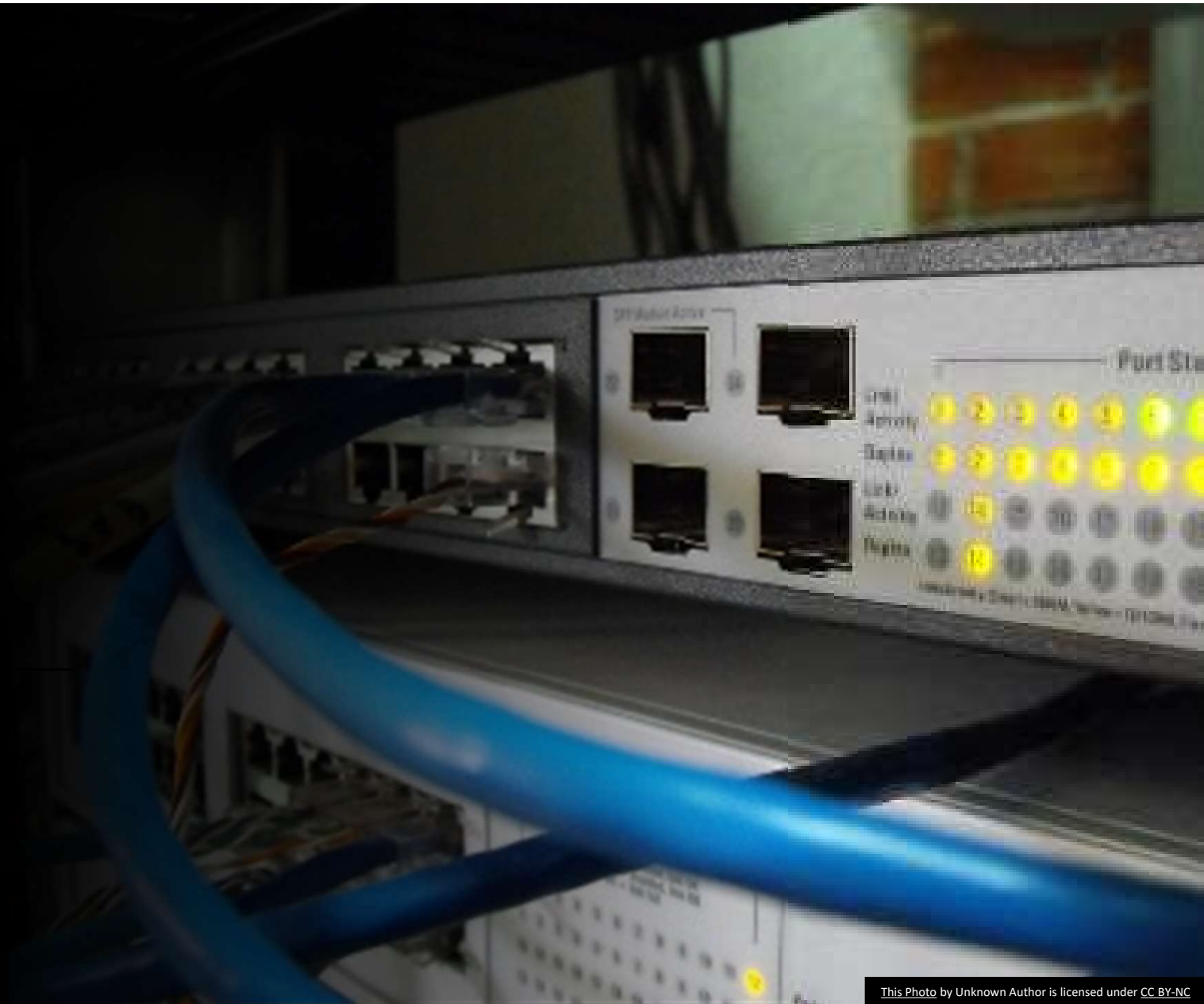




Ethernet: Switching

Mujahid Tabassum

2026



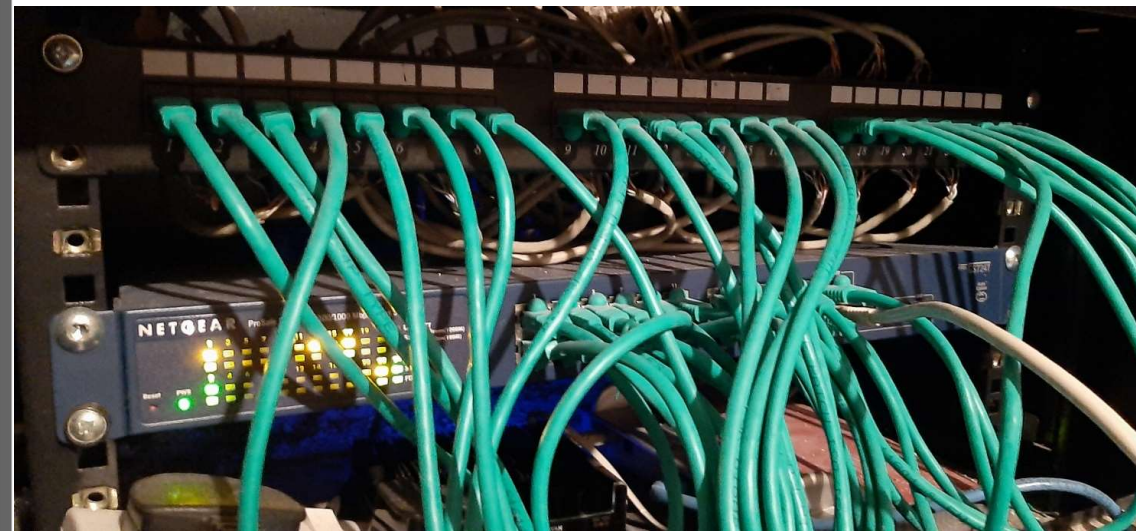
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Agenda(1st Part)

- Ethernet/MAC Addresses
- Switches
- Forwarding Table
- Frame Forwarding
- ARP

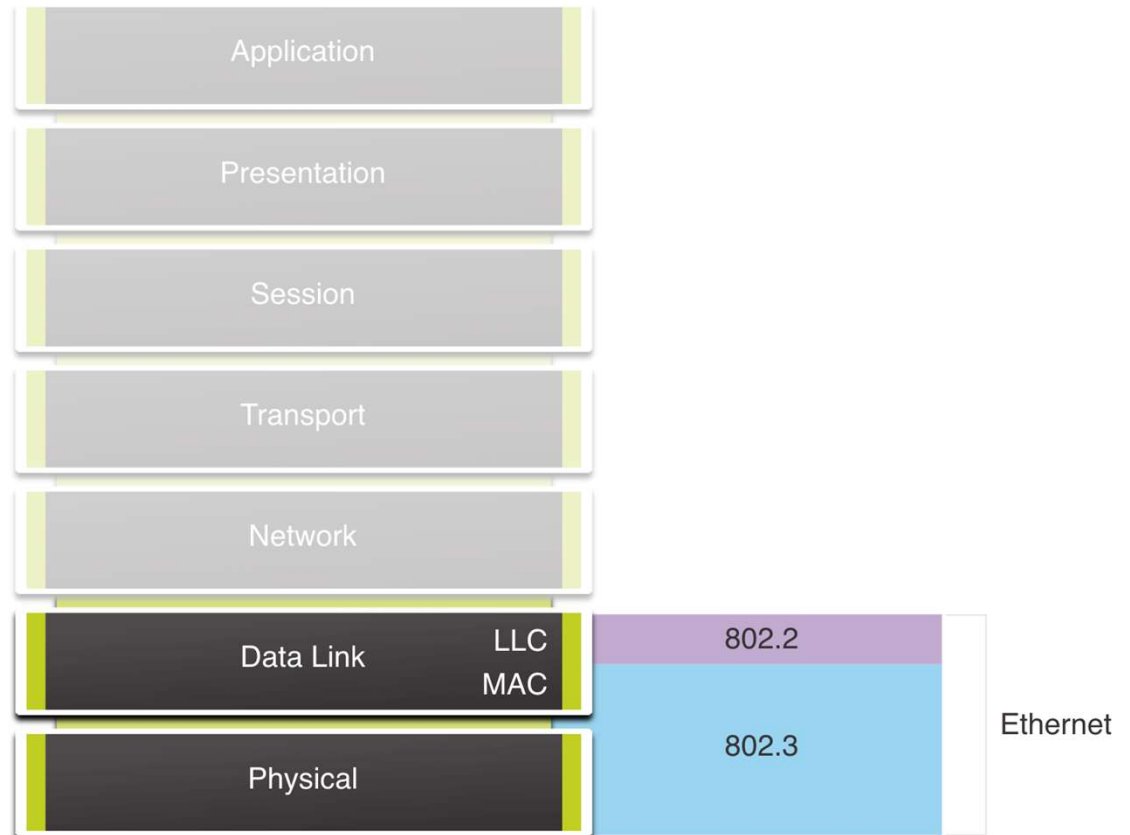
Network Switches are great!

- **A network switch forwards data packets between devices.**
- Unmanaged Switches(cheap)
 - Just plug stuff in and it starts working!
- Managed Switches(not so cheap)
 - Can also be configured for security, Virtual LANS, Traffic prioritisation...
- Better switches provide PoE (Power over Ethernet)
 - Device can use connection for power as well as communication(e.g. Networked Camera)



Ethernet Encapsulation

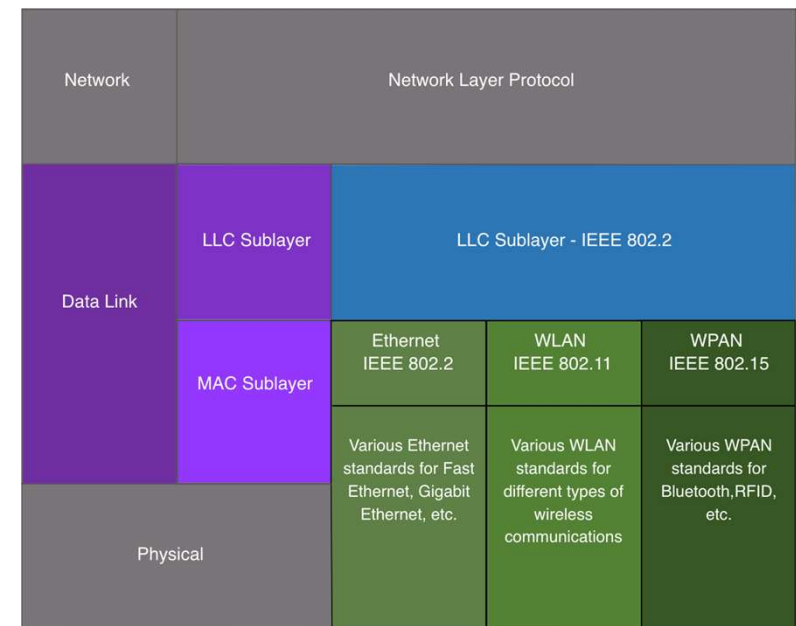
- Ethernet operates in the data link layer and the physical layer.
- It is a family of networking technologies defined in the IEEE 802.2 and 802.3 standards.



Data Link Sublayers

The 802 LAN/MAN standards, including Ethernet, use two separate sublayers of the data link layer to operate:

- **LLC Sublayer:** (IEEE 802.2) Places information in the frame to identify which network layer protocol is used for the frame.
- **MAC Sublayer:** (IEEE 802.3, 802.11, or 802.15) Responsible for data encapsulation and media access control, and provides data link layer addressing.



MAC Sublayer

The MAC sublayer is responsible for data encapsulation and accessing the media.

Data Encapsulation

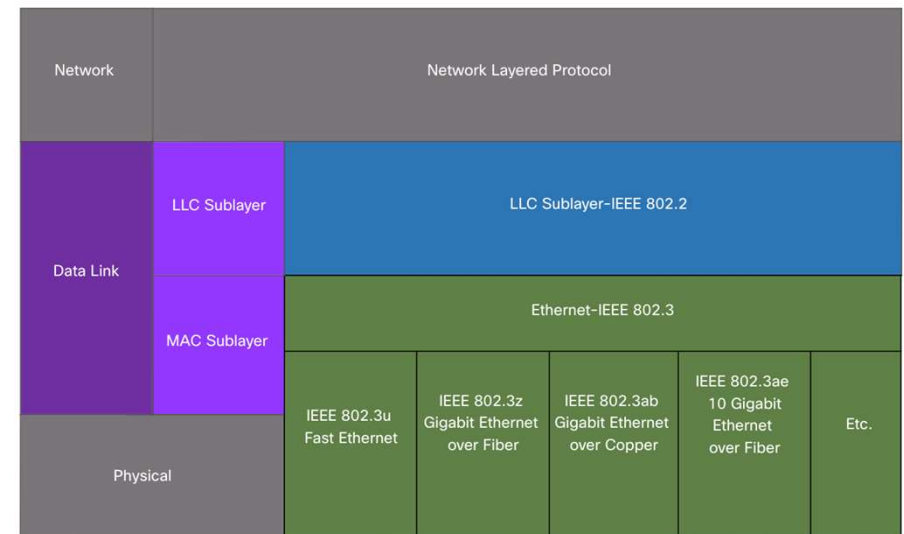
IEEE 802.3 data encapsulation includes the following:

1. **Ethernet frame** - This is the internal structure of the Ethernet frame.
2. **Ethernet Addressing** - The Ethernet frame includes both a source and destination MAC address to deliver the Ethernet frame from Ethernet NIC to Ethernet NIC on the same LAN.
3. **Ethernet Error detection** - The Ethernet frame includes a frame check sequence (FCS) trailer used for error detection.

MAC Sublayer

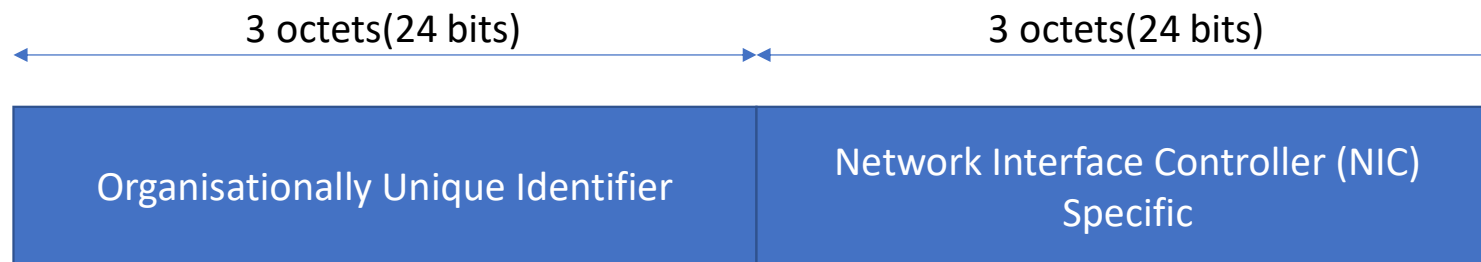
Media Access

- The IEEE 802.3 MAC sublayer includes the specifications for different Ethernet communications standards over various types of media including copper and fiber.
- Legacy Ethernet using a bus topology or hubs, is a shared, half-duplex medium. Ethernet over a half-duplex medium uses a contention-based access method, carrier sense multiple access/collision detection (CSMA/CD).
- Ethernet LANs of today use switches that operate in full-duplex. Full-duplex communications with Ethernet switches do not require access control through CSMA/CD.



Ethernet Interface: MAC Addresses

- Every Ethernet interface must have 6 byte(48 bit) MAC address
- Addresses assigned to physical interface by manufacturer/vendor



- Example: 80:19:34:95:D1:02

Usually expressed
as HEX

```
Windows IP Configuration

Host Name . . . . . : DESKTOP-QLSJ3IF
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : fritz.box

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : fritz.box
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-A4-5F-36
DHCP Enabled. . . . . : Yes
```


MAC Addresses: Hexadecimal

- An Ethernet MAC address consists of a 48-bit binary value, expressed using 12 hexadecimal values.
 - Each “hex” character represents 4 bits
- Given that 8 bits (one byte) is a common binary grouping, binary 00000000 to 11111111 can be represented in hexadecimal as the range 00 to FF,
- When using hexadecimal, leading zeroes are always displayed to complete the 8-bit representation. For example the binary value 0000 1011 is represented in hexadecimal as 0b.
- Hexadecimal numbers are often represented by the value preceded by 0x (e.g., 0x73) to distinguish between decimal and hexadecimal values in documentation.

0000	0	1000	8
0001	1	1001	9
0010	2	1010	A
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F

```
Physical Address. . . . . : 28-16-A8-65-D4-39
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 10.97.195.82(Preferred)
Subnet Mask . . . . . : 255.0.0.0
Lease Obtained. . . . . : Friday 30 September 2022 15:15:12
Lease Expires . . . . . : Saturday 1 October 2022 15:15:12
Default Gateway . . . . . : 10.128.128.128
DHCP Server . . . . . : 10.128.128.128
DNS Servers . . . . . : 10.128.128.128
NetBIOS over Tcpip. . . . . : Enabled
```

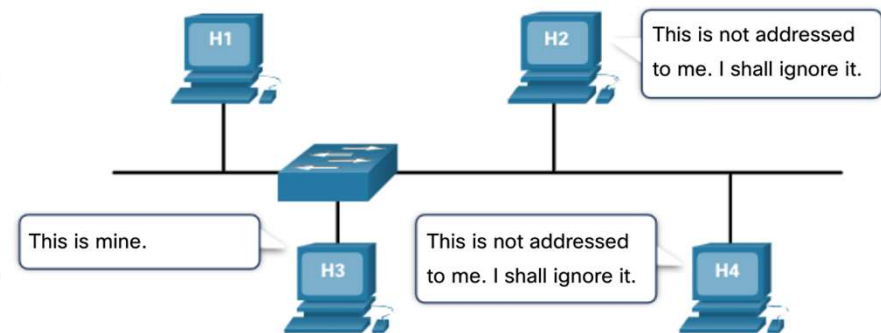
Frame Processing

- When a device is forwarding a message to an Ethernet network, the Ethernet header include a Source MAC address and a Destination MAC address.
- When a NIC receives an Ethernet frame, it examines the destination MAC address to see if it matches the physical MAC address that is stored in RAM. If there is no match, the device discards the frame. If there is a match, it passes the frame up the OSI layers, where the de-encapsulation process takes place.

Note: Ethernet NICs will also accept frames if the destination MAC address is a broadcast or a multicast group of which the host is a member.

- Any device that is the source or destination of an Ethernet frame, will have an Ethernet NIC and therefore, a MAC address. This includes workstations, servers, printers, mobile devices, and routers.

Destination Address	Source Address	Data
CC:CC:CC:CC:CC:CC	AA:AA:AA:AA:AA:AA	Encapsulated data
Frame Addressing		



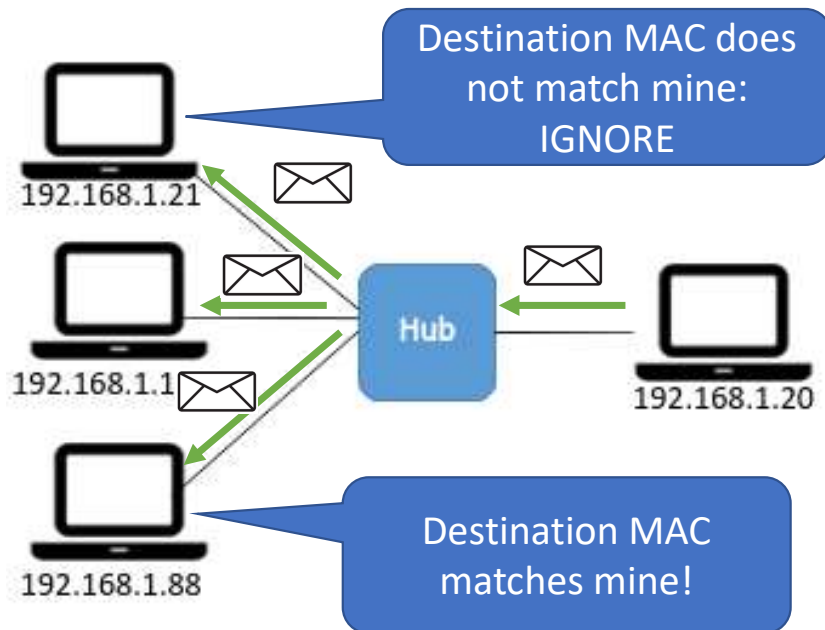
Ethernet: Frame Delivery

- Concerned with getting data to correct interface on a device.
 - Delivery of messages on a single local network
- MAC addresses are unique on the local network.
 - Represents “physical” address
- Each network interface inspects destination address of every frame. If it does not match hardware address(or broadcast address), the frame is discarded
- Once a frame is successfully received at destination, Layer 2 info is removed as the data is decapsulated and moved up the protocol stack to Network layer(layer 3).

Physical/Link Layer (layer 2) intermediate devices

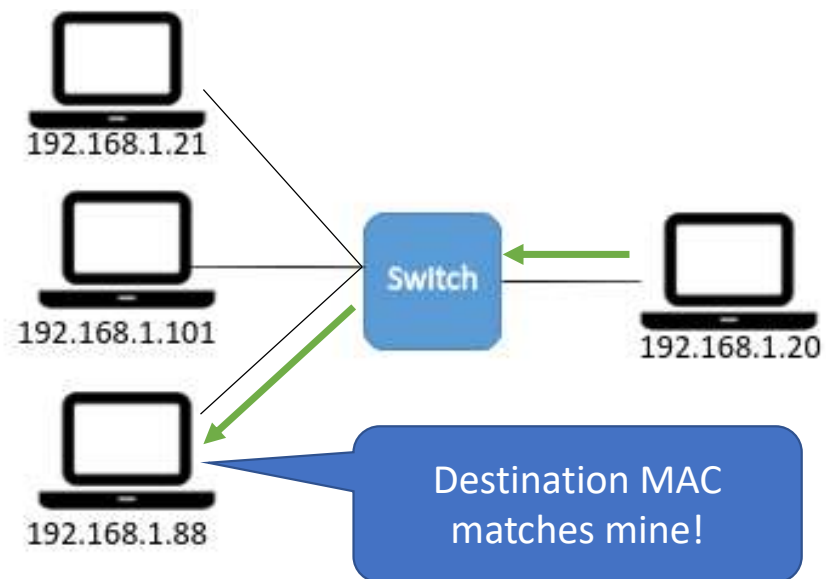
- Hub(Not cool anymore!)

- Used to connect devices
- Frames sent out on all ports.
- Shared Media (only one device can transmit)



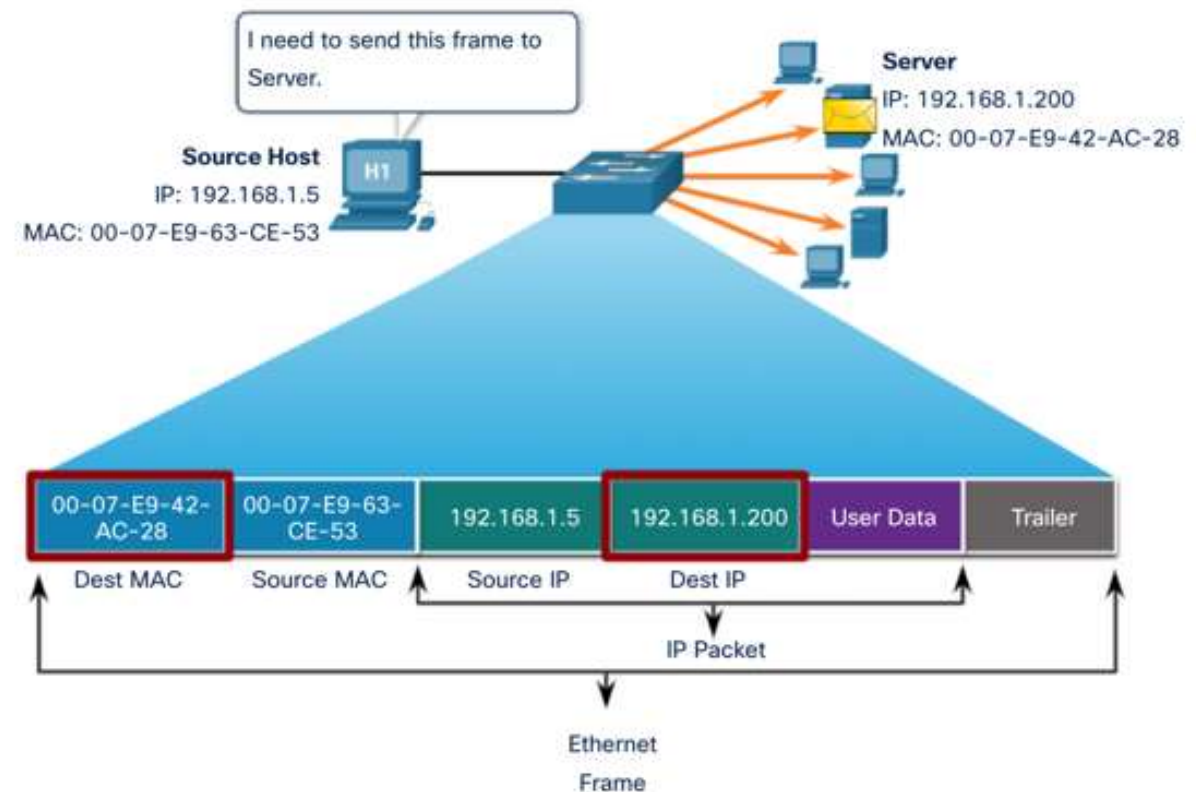
- Switch

- Replaces hubs on Ethernet networks
- Ports isolated. Frame is sent just to its proper destination(if known).



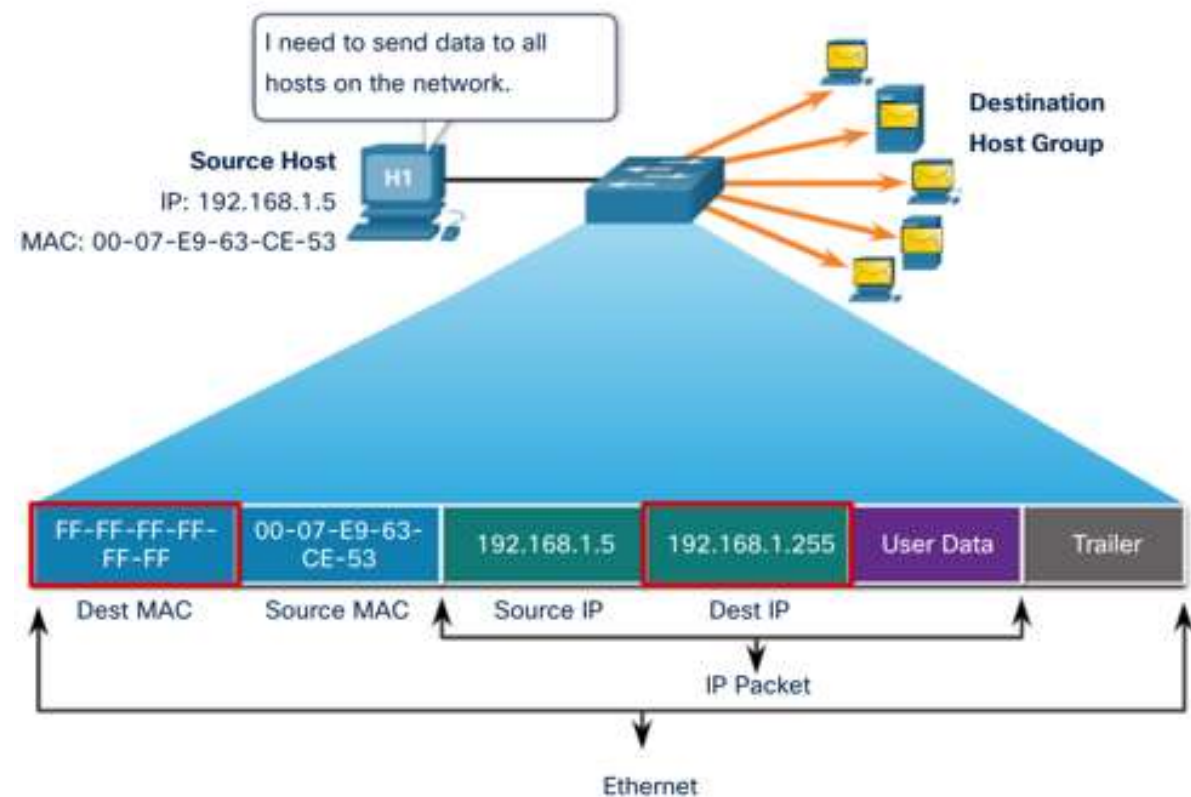
MAC Addresses: Unicast, Destination, Source

- Unicast MAC Address:
 - A unicast MAC address is the unique address that is used when a frame is sent from a single transmitting device to a single destination device.
- Destination MAC address is found using Address Resolution Protocol (ARP)
- Source MAC Address is always unicast.



MAC Addressing: Broadcast

- An Ethernet broadcast frame has a destination MAC address of FF-FF-FF-FF-FF-FF
 - received and processed by every device on the Ethernet LAN
 - Flooded on all Switch ports
- If payload is broadcast IP packet, then address portion of IP is all ones(1s)



Ethernet: Switching

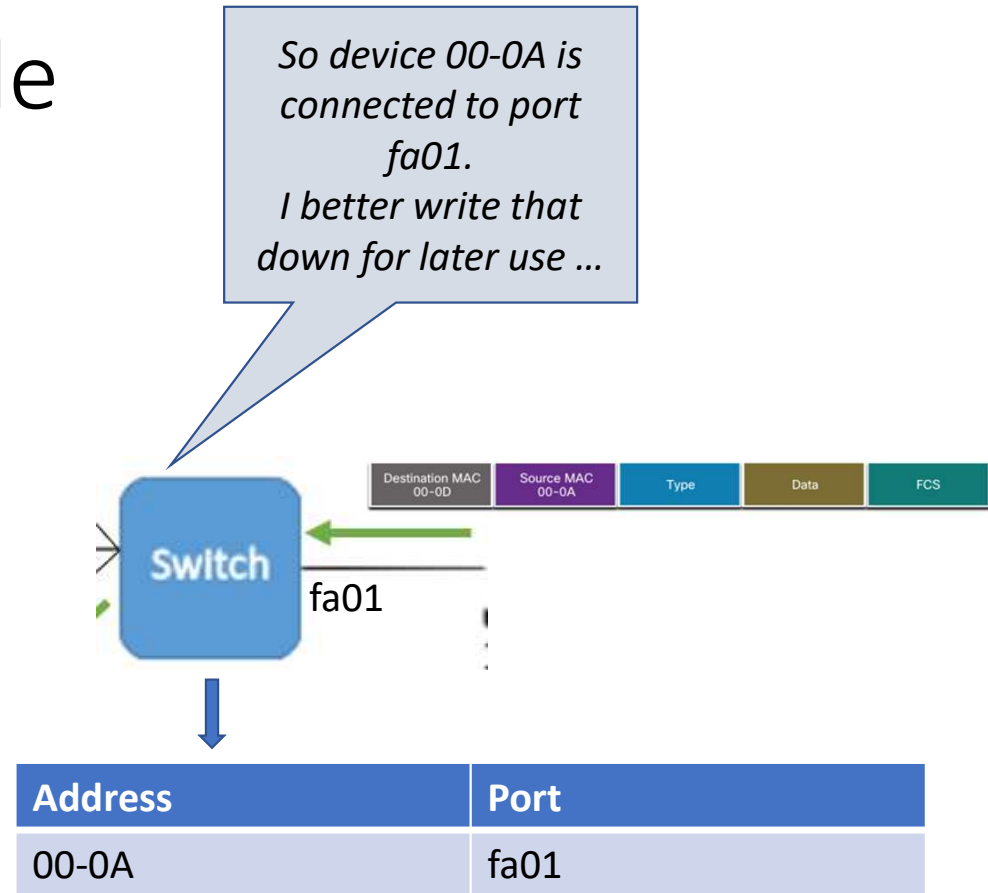
- An **Ethernet Switch** uses MAC addresses to make forwarding decisions.
- It is unaware of the data (or protocol) being carried in the data portion of the frame,
 - IPv4 packet, an ARP message.
- The switch makes its forwarding decisions based solely on the Layer 2 Ethernet MAC addresses.
- An Ethernet switch uses a **MAC address table** to make a forwarding decision for each frame
- When a switch is turned on/reset, the MAC address table is empty



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Ethernet: MAC Address Table

- A switch builds its forwarding MAC address table by examining the source MAC address of every frame that enters the switch.
 - If the source MAC address does not exist, it is added to the table along with the incoming port number. If the source MAC address does exist, the switch updates the refresh timer for that entry. By default, most Ethernet switches keep an entry in the table for 5 minutes.
- **Note: If the source MAC address does exist in the table but on a different port, the switch treats this as a new entry. The entry is replaced using the same MAC address but with the more current port number.**



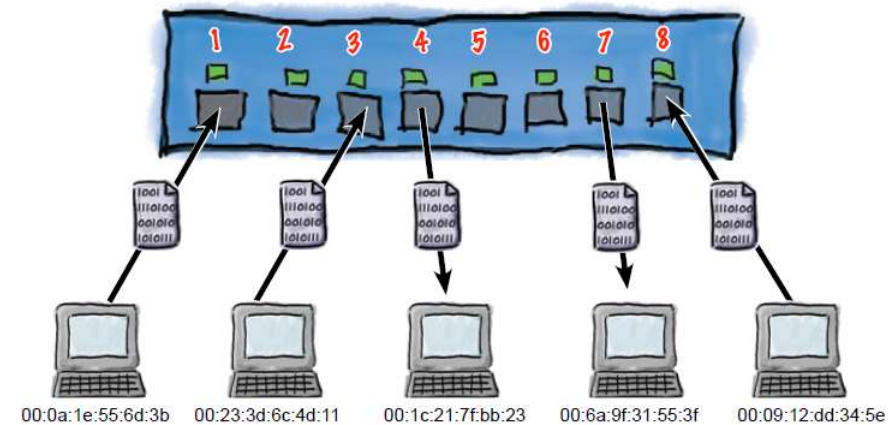
Note: MAC address in diagram abbreviated to 2 bytes

Learn

Forward

Ethernet: Switch Forwarding Table

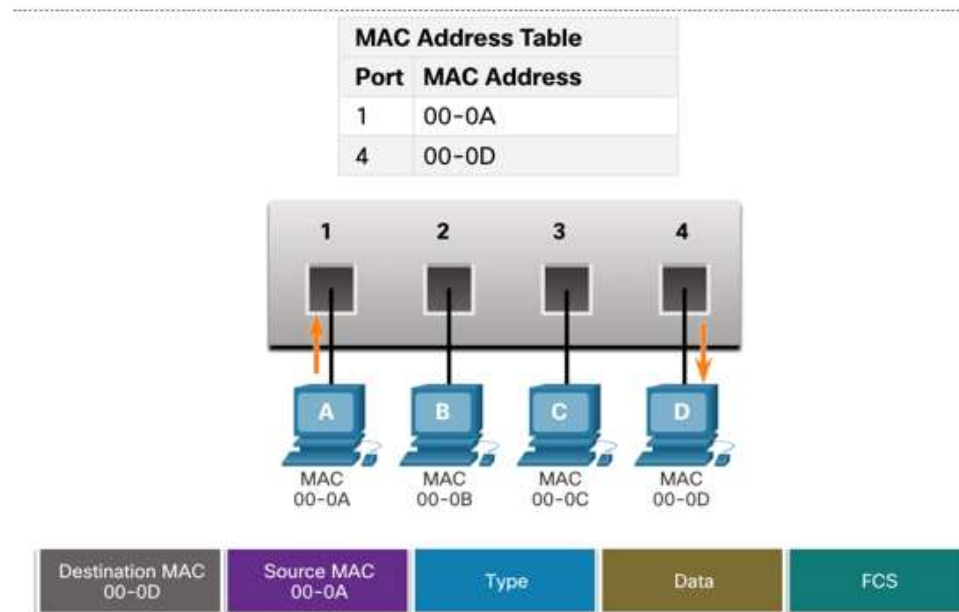
- A switch will try to match the destination MAC address of the frame and an entry in its MAC address table.
 - If the destination MAC address is in the table, it will forward the frame out the specified port.
 - If the destination MAC address is not in the table, the switch will forward the frame out all ports except the incoming port. This is called an **unknown unicast**.



MAC address	Port
00:0a:1e:55:6d:3b	1
00:23:3d:6c:4d:11	3
00:1c:21:7f:bb:23	4
00:6a:9f:31:55:3f	7
00:09:12:dd:34:5e	8

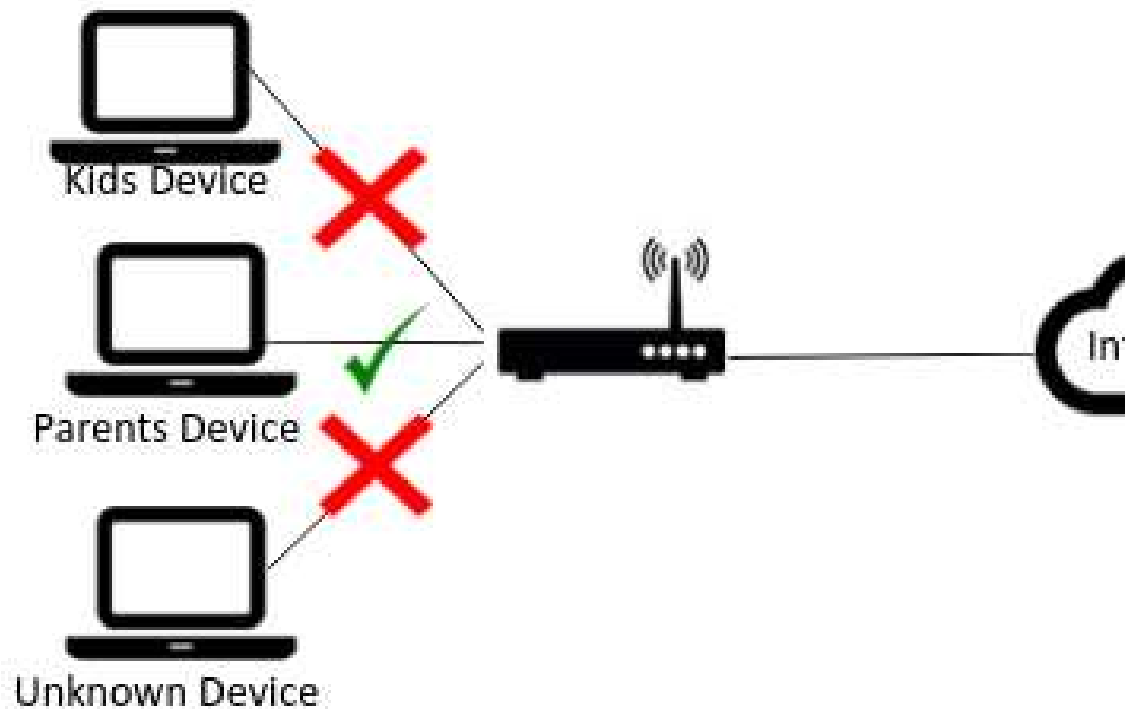
Ethernet: Switch Filtering

- When the MAC address table of the switch contains the destination MAC address, it filters the frame and forward out a single port.



Ethernet: Switch Operation Summary

- Learning
 - MAC table populated by examining traffic across ports
- Aging
 - MAC table entries are timestamped
 - Removed after a period of time
- Flooding
 - If destination MAC not in address, frame transmitted on all ports on switch
- Selective Forwarding/Filtering
 - Sending frame on one port based on MAC address
- Frame Dropping
 - Performs CRC and drops corrupted frames
 - Block frames to/from selected MAC addresses

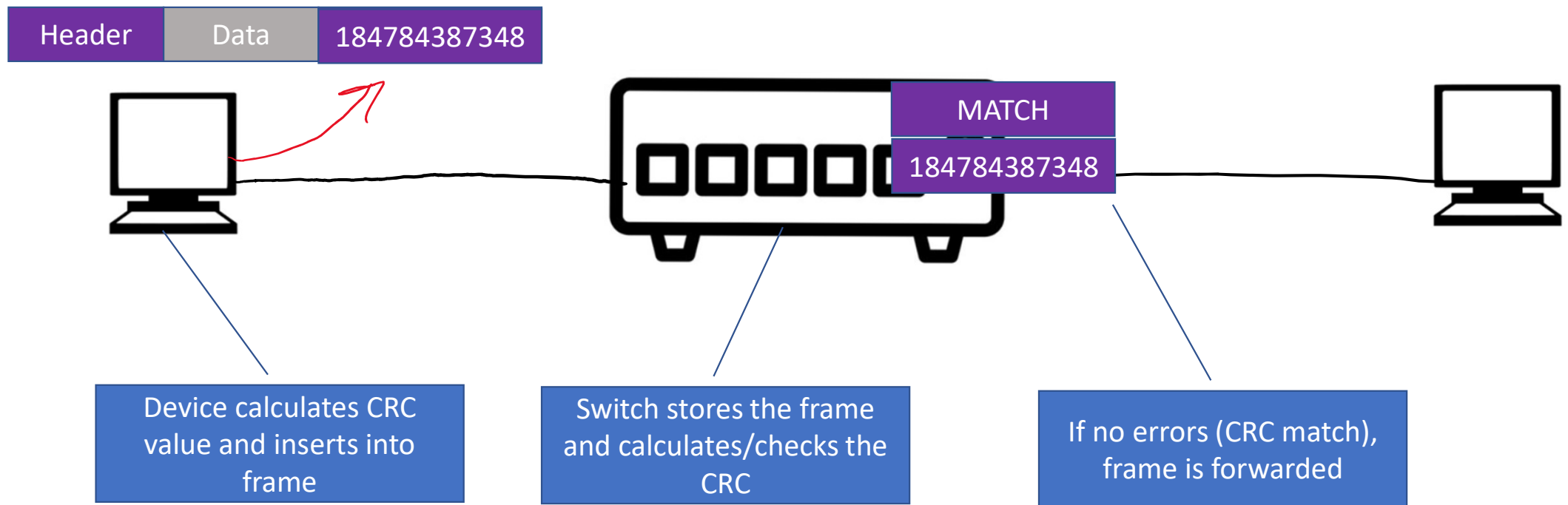


Ethernet: Switch Frame Forwarding

Ethernet: Store-and-forward

- Store-and-forward switching
 - Switch receives the entire frame and computes the CRC. If the CRC is valid, frame is forwarded as usual. If not, frame is dropped
- Advantage: High integrity. Determines if a frame has errors before propagating the frame. Discarding frames with errors reduces the amount of bandwidth consumed by corrupt data.
- Disadvantage: Overhead of CRC calculation, Added latency of process, increased frame buffering required

Store and Forward



Ethernet: Cut-Through Switching

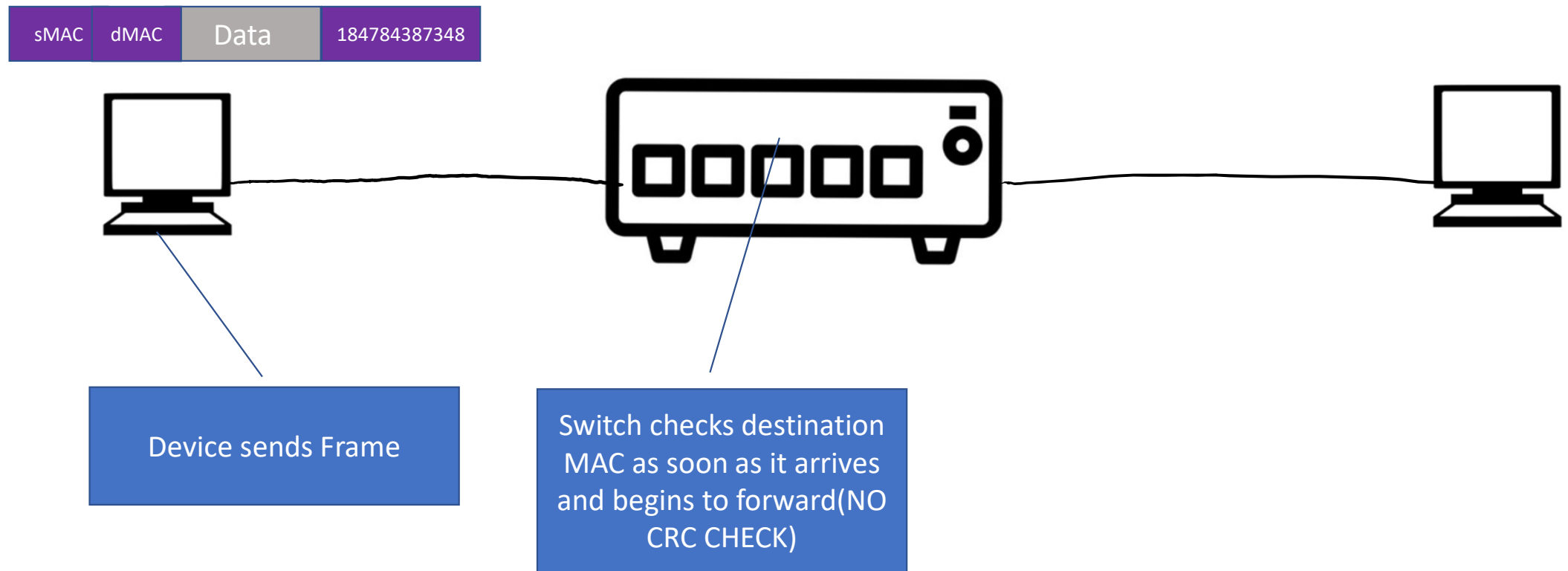
Cut-through switching acts on frame data as soon as it is received, even if the transmission is not complete.

The switch buffers just enough of the frame to read the destination MAC address and determine to which port to forward out the data.

The switch does not perform any error checking on the frame.

- **Fast-forward switching** - Offers the lowest level of latency by immediately forwarding a packet after reading the destination address.
- **Fragment-free switching** - A compromise between the high latency and high integrity of store-and-forward switching and the low latency and reduced integrity of fast-forward switching, the switch stores and performs an error check on the first 64 bytes of the frame before forwarding.

Cut Through



Ethernet: Memory Buffering

- Switches are fast, but may have to deal with large bursts of traffic/congestion, collisions, or a busy destination port
- An Ethernet switch buffer is memory that can store frames before forwarding them.
 - Supports error checking on the received frames.
 - Prevents the need to drop frames when network congestion occurs.
 - They provide a mechanism to store a frame for a particular port if autonegotiation of speed or duplex fails.
 - They hold frames when a collision occurs until normal data transmission resumes.
- **Port-Based Memory Buffering:**
 - Frames stored in Queues linked to specific ports.
- **Shared Memory:**
 - Common memory buffer for all frames.
 - Dynamic links to ports. Enables a Packet to be received on one port and transmitted on another

Port Setting

Two of the most basic settings on a switch are the bandwidth (“speed”) and duplex settings for each individual switch port. It is critical that the duplex and bandwidth settings match between the switch port and the connected devices.

There are two types of duplex settings used for communications on an Ethernet network:

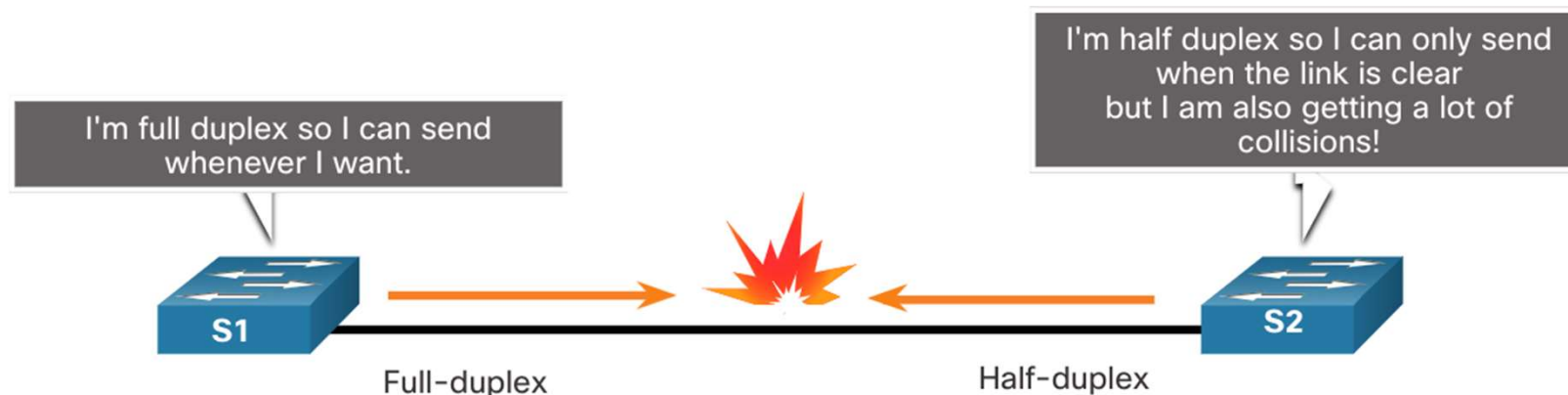
- **Full-duplex** - Both ends of the connection can send and receive simultaneously.
- **Half-duplex** - Only one end of the connection can send at a time.

Autonegotiation is an optional function found on most Ethernet switches and NICs. It enables two devices to automatically negotiate the best speed and duplex capabilities.

Note: Gigabit Ethernet ports only operate in full-duplex.

Duplex and Speed Settings

- Duplex mismatch is one of the most common causes of performance issues on 10/100 Mbps Ethernet links. It occurs when one port on the link operates at half-duplex while the other port operates at full-duplex.
- This can occur when one or both ports on a link are reset, and the autonegotiation process does not result in both link partners having the same configuration.
- It also can occur when users reconfigure one side of a link and forget to reconfigure the other. Both sides of a link should have autonegotiation on, or both sides should have it off. Best practice is to configure both Ethernet switch ports as full-duplex.

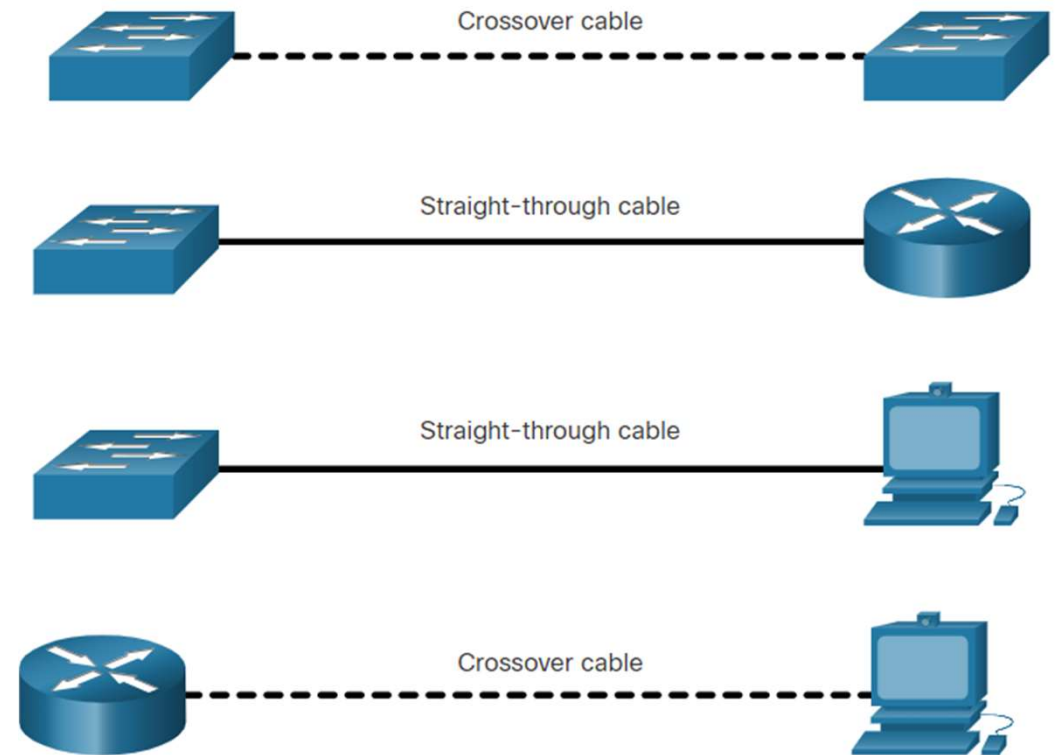


Auto-MDIX

Connections between devices once required the use of either a crossover or straight-through cable. The type of cable required depended on the type of interconnecting devices.

Note: A direct connection between a router and a host requires a cross-over connection.

- Most switch devices now support the automatic medium-dependent interface crossover (auto-MDIX) feature. When enabled, the switch automatically detects the type of cable attached to the port and configures the interfaces accordingly.
- The auto-MDIX feature is enabled by default on switches running Cisco IOS Release 12.2(18)SE or later. However, the feature could be disabled. For this reason, you should always use the correct cable type and not rely on the auto-MDIX feature.
- Auto-MDIX can be re-enabled using the **mdix auto** interface configuration command.



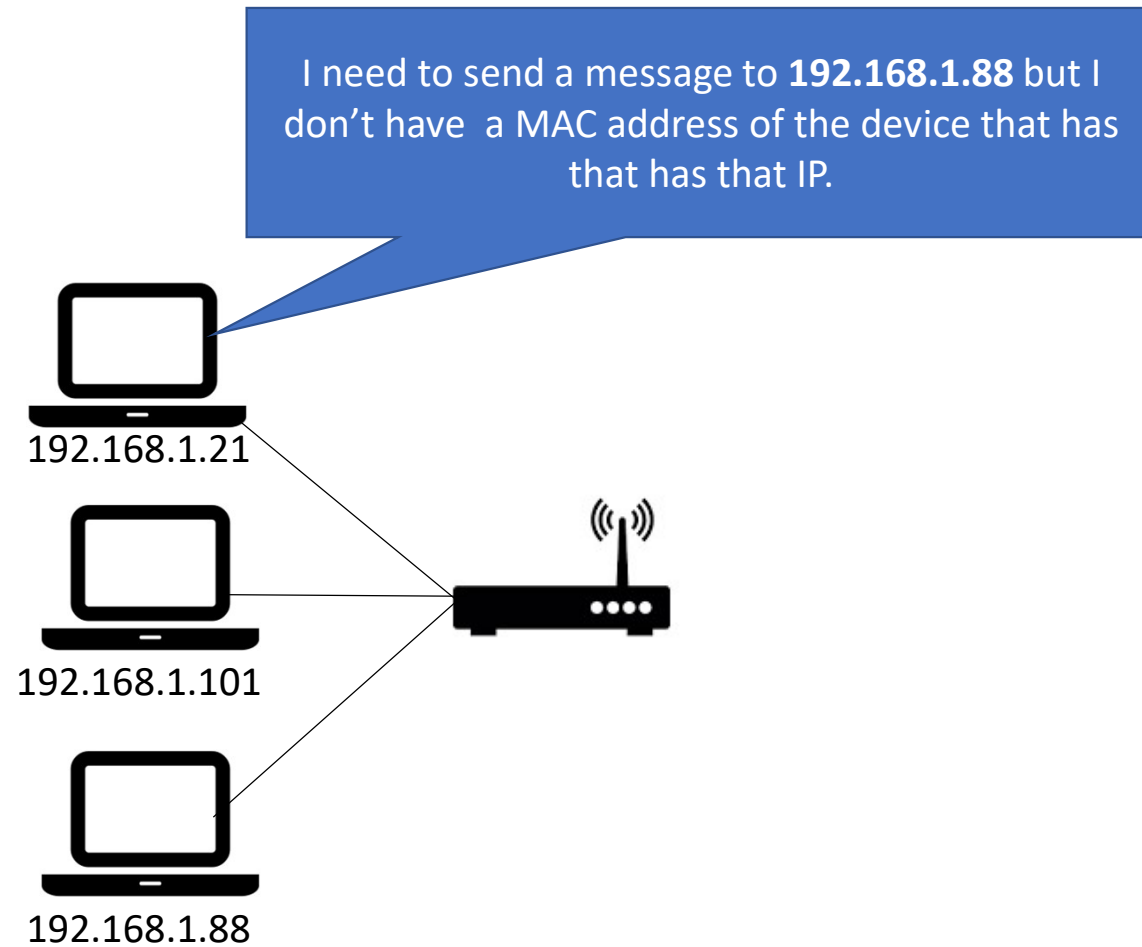
Key Points

- Data Link MAC Addresses - “Physical” address assigned to network interface.
- Switched Ethernet now standard practice (no more hubs)
- Switch Mac Address Table
 - Uses MAC addresses to Forward Ethernet Frames to Destination.

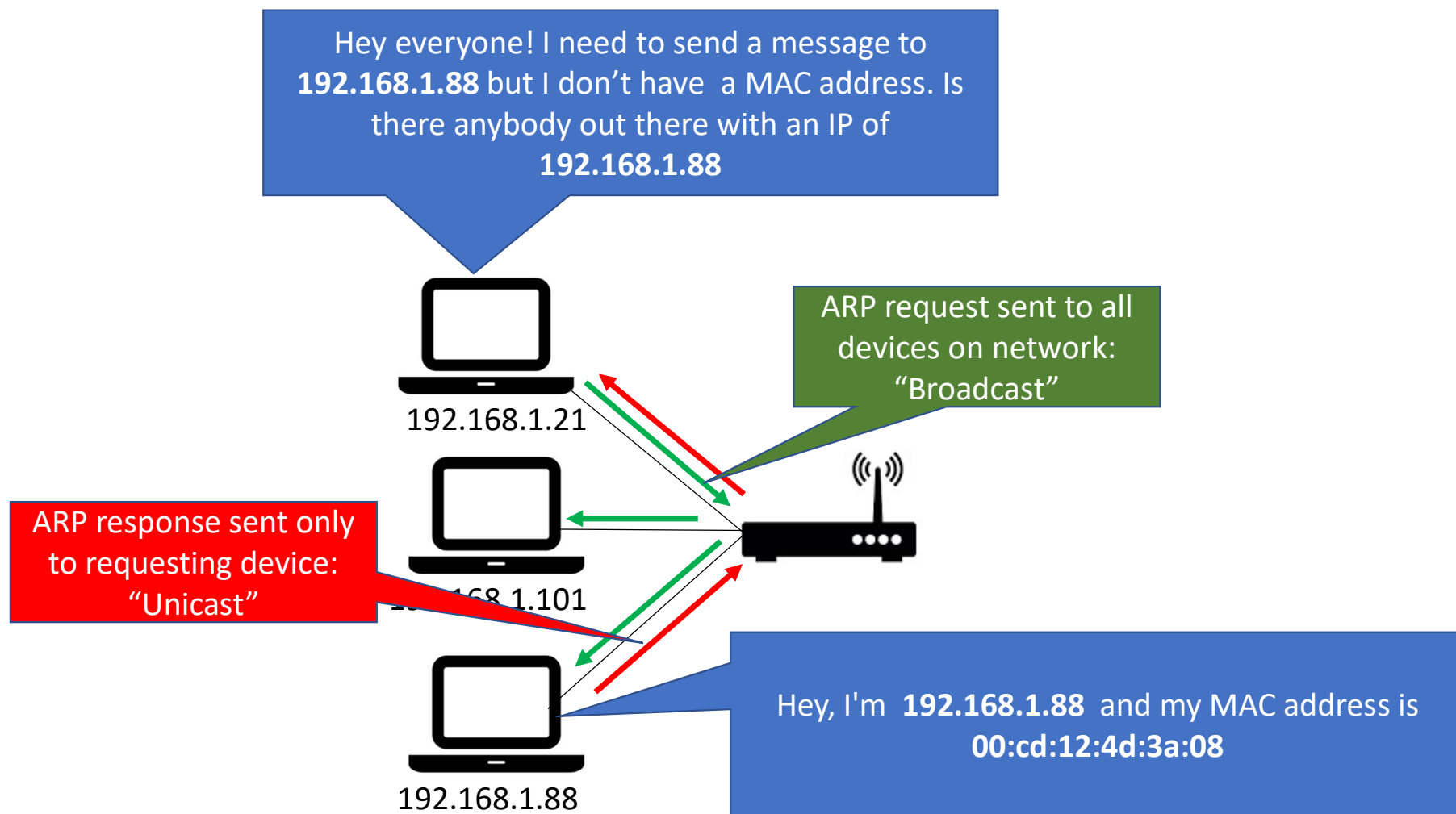
ARP

Address resolution protocol

Introduction to ARP



ARP Conversation



Address Resolution Protocol

- The *Address Resolution Protocol (ARP)* is used by a sending device when it knows the IP address of the destination but needs the Ethernet address.
- ARP is a broadcast protocol - every host on the network receives the request.
- Each host checks the request against its IP address - the right one responds.
- Hosts *remember* the hardware addresses of each other.

The ARP Process – Mapping IP to MAC Address

- Ethernet frames must have a destination MAC address
- Devices will maintain a table in memory that maps IP addresses to MAC addresses: the **ARP Table**
- The ARP table is populated using 2 mechanisms:
 - Monitor traffic on the local network
 - Broadcast an ARP request
- ARP request is broadcast to all devices on the Ethernet network.
 - Node receiving an ARP request that identifies the IP address as it sends a single response (i.e. unicast) back to the sender. Sender uses this to update the ARP table

ARP Request for device on another network

- Sending device needs to send a message to a device on another/external network. What's the destination MAC address???
- Sending device will use the MAC address of the "default gateway" - usually the MAC address of the router interface that routes to that network
- What if the ARP table doesn't contain an entry for the default gateway???
 - Device will perform ARP request for MAC

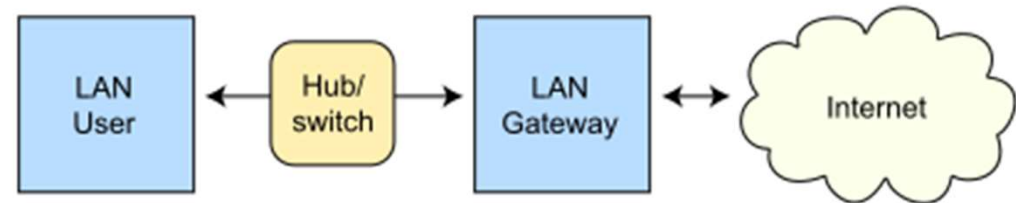
Maintaining ARP tables

- Devices remove ARP table entries that have not been used in a specified period.
 - Period differs across devices. Typically, 2 minutes for Windows.
- Example use: Device removed from network/switched off

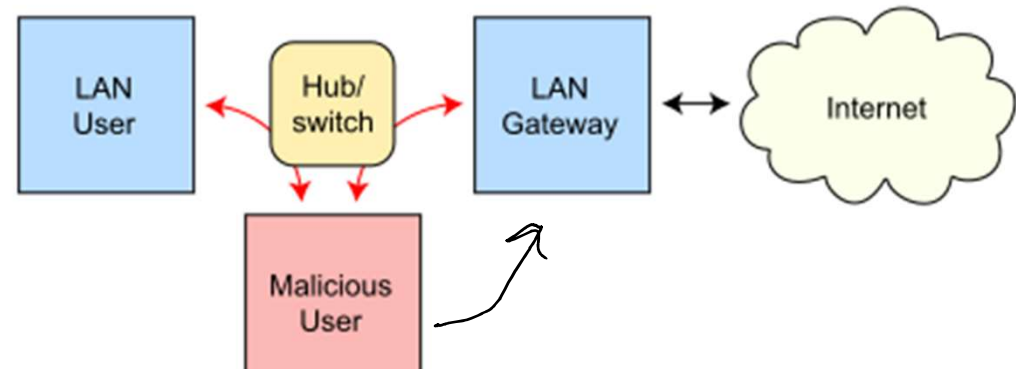
ARP Issues...

- Media Overhead
 - A lot of traffic can be generated by ARP request (broadcast). Minimal impact in typical business setting in well “segmented” network.
- Security
 - ARP Spoofing/Poisoning: Attacker forges MAC address – frames sent to wrong device...
 - Example: Using your Raspberry pi to monitor traffic from a mobile phone on your network

Routing under normal operation



Routing subject to ARP cache poisoning



Key Points -ARP

- Ethernet uses ARP to determine MAC addresses
- Each device has an IP address and a MAC address.
- ARP resolves IP addresses to MAC addresses

Quiz

- Which of the following will a Switch use to populate it's MAC-Address-Table?
 - Destination IP address
 - Source MAC address
 - Destination MAC address
 - Source IP address

Quiz

- What does a switch use to make frame forwarding decisions:
 - Destination IP address
 - Source MAC address
 - Destination MAC address
 - Source IP address